

DATA 201 – Statistical Methods in Data Science

Fall 2026 – CRN 21732

Dr. Rebin Abdulkader Muhammad

1 General Information

Instructor Information

Instructor: Dr. Rebin Abdulkader Muhammad
E-mail: Rebin.Muhammad@montgomerycollege.edu
Office Phone: 240-567-1931

Office Hours

Every day, 12:00–1:00 PM via Microsoft Teams. Students may also contact me by e-mail or Teams to arrange a meeting at another mutually convenient time.

Course and Meeting Information

Course	DATA 201 ZRA – Statistical Methods in Data Science
CRN	21732
Credits	3 semester hours
Department	Mathematics & Statistics, RV
Campus	Distance Learning
Schedule Type	Lecture
Day	Thursday
Time	6:00–8:55 PM
Location	Online – Microsoft Teams
First Scheduled Meeting	Thursday, September 3, 2026
Last Scheduled Meeting	Thursday, December 17, 2026
Official Section End Date	December 20, 2026

Course Modality: This is a synchronous online Distance Learning course. Class meetings are held live via Microsoft Teams every Thursday from 6:00–8:55 PM.

Microsoft Teams: Class meetings will be held online through the [General | Data 201-Fall2026](#) Microsoft Teams channel.

2 Course Information

Course Description

Statistical concepts and applications related to data science including advanced exploratory data analysis, nonparametric inference and simulation for larger datasets, logistic regression modeling, statistical programming, and basics of machine learning.

Prerequisite

A grade of C or better in DATA 101 or consent of department.

Course Outcomes

Upon course completion, students will be able to:

1. Select appropriate existing analytical and presentational tools for specific analyses of large databases.
2. Develop new and appropriate analytical and presentational tools for specific analyses of large databases through programming.
3. Demonstrate a competency with data science practices that allows for reproducible results.
4. Summarize findings based on complex analyses in a concise way for a general audience using multivariate graphics and statistical measures.

3 Course Materials and Technology

Textbooks and Course Resources

The primary textbook for this course is:

- Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, *An Introduction to Statistical Learning with Applications in Python* (2023). A free PDF is available from the authors' website.

Additional resources used throughout the course include:

- Andreas C. Müller and Sarah Guido, *Introduction to Machine Learning with Python*.
- Elizabeth Purdom, *Statistical Methods for Data Science*, optional supplementary resource.

Specific readings, examples, and links will be posted on Blackboard.

Technology Requirements

We will use primarily Google Colab and other open-source or free programming environments. GitHub will be used for selected homework and project submissions. Students need regular access to a Mac or PC computer and a reliable internet connection. A Chromebook is not recommended for the course work described in this syllabus. A Google account may be needed for Google Colab, and a GitHub account may be needed for course submissions.

Because this course meets synchronously through Microsoft Teams, students need access to a microphone for class participation and discussion. A webcam is also expected for interactive class activities unless the instructor indicates otherwise or an approved accommodation applies.

Computational Tools

Python and other course-approved software will be used for calculations, statistical analysis, programming, modeling, and visualization. Any restrictions on software, websites, AI tools, or other resources during an assessment will be clearly communicated in advance.

Required Online Course Resources

- **Blackboard:** Syllabus, handouts, readings, assignments, grades, announcements, and course materials.
- **Microsoft Teams:** Live class meetings, course discussions, questions and answers, peer technical assistance, and other course communication as directed.

Other Supplies

No additional required supplies are currently anticipated beyond access to the textbooks and technology/software described above. Students will be notified in advance if additional materials become necessary.

4 Course Activities and Grading

Grading

Course Activity	Weight
Project 1	15%
Project 2	15%
Lab / Homework	20%
Class Discussion	10%
Museum Visit	5%
Data Science Infographic	5%
Final Project	30%
Total	100%

Lab / Homework

Lab and homework assignments will reinforce class demonstrations, statistical modeling concepts, Python workflows, and interpretation of results. Unless otherwise announced, assignment instructions, due dates, submission locations, and required file formats will be posted on Blackboard. Selected assignments may be submitted through GitHub.

Students are responsible for confirming that their work has been successfully submitted.

Class Presentations / Discussion

This category includes short class presentations, discussion activities, problem-framing exercises, model-comparison discussions, and other activities that ask students to communicate statistical or machine-learning ideas clearly.

Late Work

Late assignments may be accepted for up to three calendar days after the deadline with a 10% penalty for each day late, unless the student has communicated with the instructor about the circumstances and an alternative submission date has been approved. Attempts at the end of the semester to submit work that was never submitted, or to resubmit earlier work solely to improve a grade, will not be accepted unless specifically authorized by the instructor.

Projects

There are two course projects and one cumulative final project. Projects are more in-depth than regular homework and will require students to work with data, build and evaluate models, interpret results, and document their workflow. Project deliverables will typically include reproducible code and a written analysis, often in the form of a Jupyter notebook or GitHub repository. Requirements, due dates, and rubrics will be posted on Blackboard.

Museum Visit

Students will complete a museum-related activity connecting data, visualization, communication, design, or interpretation to ideas from the course. Specific instructions and submission requirements will be posted on Blackboard.

Data Science Infographic

Students will create an infographic that communicates a data-science idea, model, comparison, result, or limitation to a general audience. The emphasis is on accuracy, clarity, visual communication, and effective design.

Final Project

The final project is a cumulative, end-to-end data science project. Students will frame a problem, prepare and document data, apply appropriate modeling methods, evaluate model performance, interpret results, and communicate conclusions and limitations. A final-project presentation is not required.

Make-Up Assessments

Make-up project assessments or other scheduled assessments are not automatically granted. If an unexpected, documented, or College-approved circumstance prevents you from completing an assessment as scheduled, contact the instructor as soon as possible. When a make-up is approved, the instructor will determine the appropriate format and deadline. A make-up assessment may differ from the original while measuring the same course outcomes.

Ordinary in-class participation activities generally cannot be recreated after the class meeting. Approved accommodations and College-authorized absences will be handled as required.

5 Academic Integrity and Artificial Intelligence

Academic Integrity

Honesty is the foundation of good academic work. Students are expected to submit their own work and follow Montgomery College's academic regulations and Student Code of Conduct. Collaboration is permitted only when explicitly authorized.

AI Policy for DATA 201

In this class, we recognize the value of each student's **authentic voice** and the critical-thinking skills that develop when you work through ideas and communicate them **in your own words**.

Generative artificial intelligence (AI) can be a useful learning and coding-support tool. Permitted uses may include organizing or summarizing material for study and review, requesting additional

explanations or examples, suggesting possible code, troubleshooting code, organizing a presentation, and suggesting graphics or layouts.

If AI is used in the production of code, a presentation, or a report, its use must be disclosed and cited as directed by the instructor. Students are responsible for understanding, testing, verifying, and being able to explain all submitted code and content.

AI may **not** be used to:

- submit untested or unreviewed AI-generated project code as your own work;
- produce submissions or comments for discussion assignments;
- produce unattributed or substantially unedited text for slides, reports, or project write-ups; or
- replace the student's own analysis, reasoning, or authentic written voice.

Written work, calculations, analysis, and explanations submitted for evaluation must represent the student's own understanding. Unauthorized or undisclosed AI-generated work may be treated as a violation of academic integrity.

6 Course Communication

Montgomery College e-mail is the official means of College communication. Blackboard announcements will be used for course announcements, schedule updates, assignments, and reminders. Microsoft Teams will be used for synchronous class meetings and may also be used for questions, course discussions, feedback, and technical assistance.

7 Attendance and Participation Expectations

Learning is an active and social process. Students should attend all scheduled Teams meetings and arrive on time. Being present, active, and engaged provides the strongest opportunity to learn the material and participate in demonstrations, discussions, and collaborative activities.

Strong participation includes completing or attempting assigned preparation, taking notes, following and running code during demonstrations, asking relevant questions, contributing to discussions, and remaining focused on course activities.

If circumstances are interfering with your ability to attend regularly, contact the instructor as soon as possible. It is the student's responsibility to officially drop or withdraw from the course when necessary; non-attendance does not constitute official withdrawal.

8 Tentative Fall 2026 Schedule

The instructor reserves the right to alter the schedule and topics according to the progression and needs of the class. Course materials and schedule updates will be posted on Blackboard.

Week	Date	Main Topic	Primary Resource / Assignment
1	Sep. 3	Machine learning & statistical learning foundations: inference vs. prediction, supervised vs. unsupervised learning, regression vs. classification, problem framing, tradeoffs	ISLP Ch. 1–2; Google Colab introduction; short problem-framing activity
2	Sep. 10	Data preparation and reproducible workflows: data quality, missing values, categorical variables, scaling, transformations, leakage	ISLP Ch. 2 lab; Müller & Guido Ch. 4; preprocessing lab
3	Sep. 17	Logistic regression and classification foundations: probabilities, decision boundaries, classification vs. inference, loss/objective ideas	ISLP Ch. 4; logistic regression Colab lab
4	Sep. 24	Classification algorithms: logistic regression, LDA, QDA, KNN, model comparison	ISLP Ch. 4; Müller & Guido Ch. 2; classifier comparison lab
5	Oct. 1	Model evaluation: train/test split, confusion matrix, accuracy, precision, recall, specificity, F1, ROC-AUC, thresholds	ISLP Ch. 5; Müller & Guido Ch. 5; evaluation metrics lab
6	Oct. 8	Cross-validation, overfitting, underfitting, hyperparameter tuning, model selection	ISLP Ch. 5; Müller & Guido Ch. 5; Project 1 due
7	Oct. 15	Feature engineering and modeling pipelines: encoding, scaling, transformations, feature selection, reproducible scikit-learn pipelines	Müller & Guido Ch. 3, 4, and 6; pipeline-building lab
8	Oct. 22	Regularization and model complexity: ridge, lasso, bias–variance tradeoff, interpretability vs. flexibility	ISLP Ch. 6; regularization lab
9	Oct. 29	Decision trees: splitting, impurity, pruning, classification and regression trees, interpretation	ISLP Ch. 8; Müller & Guido Ch. 2; decision-tree lab
10	Nov. 5	Ensemble methods: bagging, random forests, boosting, variable importance	ISLP Ch. 8; Müller & Guido Ch. 2; ensemble comparison lab
11	Nov. 12	End-to-end model comparison: pipelines, tuning, comparing logistic/regularized models with trees and ensembles	Integrated ISLP + Müller & Guido workflow; Project 2 due
12	Nov. 19	Neural networks: perceptron, hidden layers, activation functions, loss functions, gradient descent, backpropagation intuition	ISLP Ch. 10; Müller & Guido neural-network section; introductory NN lab

–	Nov. 26	Fall Break – College Closed; No Class	–
13	Dec. 3	Introductory deep learning: deeper networks, regularization, validation, practical applications	ISLP Ch. 10; deep-learning Colab lab / discussion
14	Dec. 10	Responsible and reproducible ML: bias, data quality, drift, security, accountability, model limitations	Selected textbook examples; model-risk discussion; Final Project workshop
15	Dec. 17	Final Project	Final Project due

9 Important Fall 2026 Dates

- **Fall classes begin:** August 31, 2026
- **First scheduled DATA 201 meeting:** Thursday, September 3, 2026
- **Labor Day – College closed:** September 7, 2026
- **MC Day – no classes for students:** October 20, 2026
- **Fall Break – College closed:** November 25–29, 2026
- **Final Exam Week:** December 14–20, 2026
- **Last scheduled DATA 201 meeting:** Thursday, December 17, 2026
- **Official end of semester/section:** December 20, 2026

DATA 201.21732 Class Drop Dates

- **Refund Date:** September 9, 2026
- **No Grade / Change to Audit Deadline:** September 23, 2026
- **Last Date to Withdraw:** November 18, 2026

It is the student's responsibility to officially drop or withdraw from the course. Non-attendance does not constitute official withdrawal.

For the full College calendar, see the [Academic Year Calendar 2026–2027](#).

10 Student Resources and College Policies

Disability Support Services

Students who may need an accommodation due to a disability should contact Disability Support Services (DSS). Approved accommodations will be provided in accordance with the DSS accommodation letter.

<https://www.montgomerycollege.edu/counseling-and-advising/disability-support-services/index.html>

Learning Centers

Montgomery College Learning Centers provide academic support and tutoring resources to enrolled students.

<https://www.montgomerycollege.edu/academics/support/learning-centers/index.html>

Counseling and Advising

<https://www.montgomerycollege.edu/counseling-and-advising/>

Assessment and Testing Centers

<https://www.montgomerycollege.edu/admissions-registration/assessment-testing-centers/>

Student Forms

<https://www.montgomerycollege.edu/admissions-registration/student-resources/student-forms.html>

Public Safety

<https://www.montgomerycollege.edu/life-at-mc/public-safety/index.html>

Pregnant and Parenting Students

Montgomery College provides Title IX protections and academic support for students who are pregnant or experiencing pregnancy-related conditions. Medically necessary absences will be excused and missed work may be made up as required by College policy. See [Pregnant and Parenting Students](#) for details.

Sexual Misconduct / Title IX

<https://www.montgomerycollege.edu/policies-and-procedures/title-ix/sexual-misconduct-policy-and-procedures.html>

11 Inclusivity and Commitment to Student Success

Your well-being and success in this course matter. Our classroom should be an inclusive learning environment in which students can ask questions, make mistakes, exchange ideas respectfully,

and contribute their experiences and perspectives. Course materials and activities should support respectful engagement across differences. If an assignment due date conflicts with a religious observance, please contact the instructor as early as possible so that appropriate arrangements can be considered.

12 Syllabus Changes

This syllabus and tentative schedule may be adjusted when necessary to support the progress of the course. Changes will be communicated through Blackboard.